Model, Prompts & Configuration

1. LLM Configuration

The system uses the Google Gemini API (via google-genai SDK) for generating answers and performing architectural analysis from the retrieved AUTOSAR HLD context.

1. Current Provider: Google AI Studio
2. Alternative Provider: Vertex AI
3. Model: gemini-2.5-flash
4. API Interface: Google GenAI Python SDK
5. Purpose: Generate document-grounded responses, extract AUTOSAR components, ports, interfaces, signals, and identify architectural inconsistencies.
6. API Key: Stored securely through environment variables (.env) and excluded from version control.

The LLM is called only after relevant document chunks have been retrieved from FAISS and passed through the relevance filtering stage.

2. Embedding Configuration

Embedding Framework: Google GenAI Embeddings

Embedding Model: gemini-embedding-001

The embedding model converts both document chunks and user questions into 3072-dimensional numerical vector representations. These vectors are L2-normalized and used for cosine semantic similarity search.

The model is loaded dynamically when processing uploads and user queries.

3. Vector Store / Knowledge Base

The project uses FAISS (Facebook AI Similarity Search) as a persistent vector store.

1. Vector Store: FAISS (IndexFlatIP with L2 Normalization)
2. Metadata: Python Pickle / SQLite DB
3. Storage Path: vector_store/doc_{id}.faiss

The FAISS index stores document embeddings, while metadata maintains information such as:

1. Document name & Document ID
2. Page number (exact source citation)
3. Chunk text & Chunk ID

This metadata enables the system to provide accurate, grounded document and page-level citations with every generated answer.

4. RAG Configuration

Parameter	Value
Embedding Model	gemini-embedding-001
Top-K Retrieval	6
Chunk Size	800 characters
Chunk Overlap	150 characters
Vector Store	FAISS (IndexFlatIP)
Similarity Metric	Cosine Similarity (Normalized IP)
Storage Engine	Local Vector Store + SQLite

5. RAG System Prompt
You are an AUTOSAR HLD document analysis assistant.
Answer questions ONLY based on the provided document context.
If the answer is not in the context, say so clearly.
Always cite the page numbers where information was found.
Be precise and technical.

If the provided context does not contain sufficient information to answer the question, respond:
"This information is not found in the provided document sections."

Keep the response concise, technically rigorous, and grounded strictly in the retrieved AUTOSAR HLD document context.

6. Citation Configuration
Citations are generated dynamically from the page metadata associated with the retrieved FAISS chunks.
The system provides: Document name, Specific Page Number(s)
Only sources associated with the top-ranked relevant chunks are cited. Redundant citations from multiple chunks on the identical page are automatically deduplicated.
This guarantees that all displayed source tags strictly correspond to the technical sections used to formulate the engineering response.

7. Environment Configuration